PRODUCT USER MANUAL

OMI Ocean Health Chl OceanColour

OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_trend
OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_area_averaged_mean
OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_trend
OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_area_averaged_mean
OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_trend
OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_area_averaged_mean

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GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
OC	Ocean Colour
MED	Mediterranean Sea
BLK	Black Sea
BAL	Baltic Sea

Table of Content

GLOSSARY AND ABBREVIATIONS **3**

DATA ACCESS **5**

I INTRODUCTION **5**

I.1. Summary **5**

I.2. History of changes **6**

II DESCRIPTION OF THE PRODUCT **7**

II.1. General information **7**

II.1.1. Mediterranean Sea 7

II.1.1. Black Sea 7

II.1.1. Baltic Sea 8

II.2. Details of the datasets **10**

II.2.1. Mediterranean Sea 10

II.2.2. Black Sea 11

II.2.3. Baltic Sea 12

II.3. Product System Description **13**

II.3.1. Ocean Colour Trend 13

II.3.2. Ocean Colour Monthly Mean Time Series 13

III FILE FORMAT **13**

IV FILES NOMENCLATURE **15**

IV.1. Nomenclature of files **15**

V REFERENCES **15**

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download. Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Ocean Monitoring Indicator (OMI). It provides 1D (time) time series and 2D (latitude and longitude) data of chlorophyll concentration trends for the period 1997-2023 for Mediterranean Sea, Black Sea and Baltic Sea.

The document refers to these products:

OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_trend
OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_area_averaged_mean
OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_trend
OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_area_averaged_mean
OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_trend
OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_area_averaged_mean

These data are computed using MY [1] data which provide chlorophyll concentrations from satellite observations. The products are published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The products are available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.
MY data are described in detail in the corresponding Quality Information Document (QUID) (MED and BLK: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OC-QUID-009-141to144-151to154.pdf; BAL: CMEMS-OC-QUID-009-131to134.pdf (copernicus.eu)).

OMI products are detailed in the corresponding Quality Information Documents (QUID):

[1] OCEANCOLOUR_MED_BGC_L4_MY_009_144, OCEANCOLOUR_BLK_BGC_L4_MY_009_154, OCEANCOLOUR_BAL_BGC_L3_MY_009_134

MEDSEA: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-HEALTH-CHL-MEDSEA-OCEANCOLOUR.pdf;

BLKSEA: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-HEALTH-CHL-BLKSEA-OCEANCOLOUR.pdf;

BALTIC: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-HEALTH-CHL-BALTIC-OCEANCOLOUR.pdf;

I.2. History of changes

2021/01	Creation of the document. The content has been extracted and updated from the former OCTAC OMI PUM v4.0
2021/12	Update for EIS December 2021: • Temporal extension to the end of 2020
2022/11	Update for EIS November 2022: • Temporal extension to the end of 2021
2023/11	Update for EIS November 2023: Temporal extension to the end of 2022
2024/11	Update for EIS November 2024: Temporal extension to the end of 2023

II DESCRIPTION OF THE PRODUCT

II.1. General information

II.1.1. Mediterranean Sea

Product(s) name	OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_trend OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_area_averaged_mean
Geographical coverage	MED (30°-46°N, -6°-36.5°E)
Datasets	omi_health_chl_medsea_oceanclour_trend omi_health_chl_medsea_oceanclour_area_averaged_mean
Update frequency	Annual
Available time series	From 1997 to year -1
Delivery mechanism	Marine Data Store
Temporal/Spatial resolution	Monthly time series, 20-year trend
Integration depth	N/A
Format	NetCDF CF1.4

II.1.1. Black Sea

Product(s) name	OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_trend OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_area_averaged_mean
Geographical coverage	BLK (40°-48°N,26.5°-42°E)
Datasets	omi_health_chl_blksea_oceanclour_trend

	omi_health_chl_blksea_oceanclour_area_averaged_mean
Update frequency	Annual
Available time series	From 1997 to year -1
Delivery mechanism	Marine Data Store
Temporal/Spatial resolution	Monthly time series, 20-year trend
Integration depth	N/A
Format	NetCDF CF1.4

II.1.1. Baltic Sea

Product(s) name	OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_trend OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_area_averaged_mean
Geographical coverage	BAL (53.25°-65.85°N, 9.25°-30°E)
Datasets	omi_health_chl_baltic_oceanclour_trend omi_health_chl_baltic_oceanclour_area_averaged_mean
Update frequency	Annual
Available time series	From 1997 to year -1
Delivery mechanism	Marine Data Store
Temporal/Spatial resolution	Monthly time series, 20-year trend

Integration depth	N/A
Format	NetCDF CF1.4

II.2. Details of the datasets

II.2.1. Mediterranean Sea

OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_trend	
omi_health_chl_medsea_oceancolour_trend: contains the 2D Mediterranean Sea chlorophyll trend	
Variables name in the NetCDF file	TREND
Long_name	Change in chlorophyll concentration in seawater over time
Standard_name	change_in_mass_concentration_of_chlorophyll_in_seawater_over_time
Units	% yr-1

OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_area_averaged_mean	
omi_health_chl_medsea_oceancolour_area_averaged_mean: contains the 1D Mediterranean Sea chlorophyll monthly mean time series	
Variables name in the NetCDF file	CHLOR_A
Long_name	Monthly mean concentration of chlorophyll over time
Standard_name	mass_concentration_of_chlorophyll_in_seawater
Units	mg m-3
Variables name in the NetCDF file	CHLOR_DESEASONALIZED
Long_name	Monthly mean concentration of deseasonalized chlorophyll over time
Standard_name	mass_concentration_of_chlorophyll_in_seawater
Units	mg m-3

Note that the standard name and units of trend in **OMI_HEALTH_CHL_MEDSEA_OCEANCOLOUR_trend** product are not yet included in CF conventions.

II.2.2. Black Sea

OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_trend	
omi_health_chl_blksea_oceancolour_trend: contains the 2D Black Sea chlorophyll trend	
Variables name in the NetCDF file	TREND
Long_name	Change in chlorophyll concentration in seawater over time
Standard_name	change_in_mass_concentration_of_chlorophyll_in_seawater_over_time
Units	% yr-1

OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_area_averaged_mean	
omi_health_chl_blksea_oceancolour_area_averaged_mean: contains the 1D Black Sea chlorophyll monthly mean time series	
Variables name in the NetCDF file	CHLOR_A
Long_name	Monthly mean concentration of chlorophyll over time
Standard_name	mass_concentration_of_chlorophyll_in_seawater
Units	mg m-3
Variables name in the NetCDF file	CHLOR_DESEASONALIZED
Long_name	Monthly mean concentration of deseasonalized chlorophyll over time
Standard_name	mass_concentration_of_chlorophyll_in_seawater
Units	mg m-3

Note that the standard name and units of trend in **OMI_HEALTH_CHL_BLKSEA_OCEANCOLOUR_trend** product are not yet included in CF conventions.

II.2.3. Baltic Sea

OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_trend	
omi_health_chl_baltic_oceancolour_trend: contains the 2D Baltic Sea chlorophyll trend	
Variables name in the NetCDF file	TREND
Long_name	Change in chlorophyll concentration in seawater over time
Standard_name	change_in_mass_concentration_of_chlorophyll_in_seawater_over_time
Units	% yr-1

OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_area_averaged_mean	
omi_health_chl_baltic_oceancolour_area_averaged_mean: contains the 1D Baltic Sea chlorophyll monthly mean time series	
Variables name in the NetCDF file	CHLOR_A
Long_name	Monthly mean concentration of chlorophyll over time
Standard_name	mass_concentration_of_chlorophyll_in_seawater
Units	mg m-3
Variables name in the NetCDF file	CHLOR_DESEASONALIZED
Long_name	Monthly mean concentration of deseasonalized chlorophyll over time

Standard_name	mass_concentration_of_chlorophyll_in_seawater
Units	mg m-3

Note that the standard name and units of trend in **OMI_HEALTH_CHL_BALTIC_OCEANCOLOUR_trend** product are not yet included in CF conventions.

II.3. Product System Description

II.3.1. Ocean Colour Trend

MY[2] data, which provide the regional chlorophyll concentration, are used to compute the trend maps. Trend maps were obtained by using the methodology described in Colella et al. (2016), where the Mann-Kendall test (Mann, 1945; Kendall, 1975) and the Sens's method (Sen, 1968) are applied on deseasonalized monthly time series, as obtained from the X-11 technique, to estimate, trend magnitude and its significance. The trend is expressed in % year^{-1} that represents the relative changes (i.e., percentage) corresponding to the dimensional trend [mg m^{-3} y^{-1}] with respect to the chlorophyll climatology (1997-2014).

II.3.2. Ocean Colour Monthly Mean Time Series

MY[3] data, which provide the regional chlorophyll concentration, are used to compute the monthly time series and to derive the seasonal cycle. The deseasonalized time series is obtained by applying the X-11 seasonal adjustment methodology on the original time series as described in Colella et al. (2016), and then Mann-Kendall test and Sens's method are subsequently applied to obtain the magnitude of trend.

III FILE FORMAT

[2]OCEANCOLOUR_MED_BGC_L4_MY_009_144, OCEANCOLOUR_BLK_BGC_L4_MY_009_154, OCEANCOLOUR_BAL_BGC_L3_MY_009_134

[3] OCEANCOLOUR_MED_BGC_L4_MY_009_144, OCEANCOLOUR_BLK_BGC_L4_MY_009_154, OCEANCOLOUR_BAL_BGC_L3_MY_009_134

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1. Nomenclature of files

OMI files nomenclature when downloaded through the CMEMS Marine Data Store Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference;
- **<timestep>** is the first time step contained in the file [YYYYMMDD];
- **<productiontime>** is the production time when the file was generated [YYYYMMDD];

Example:

omi_health_chl_medsea_oceancolour_trend_19970901_P20220503.nc

V REFERENCES

See the Quality Information Document (QUID):

http://marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-HEALTH-CHL-MEDSEA-OCEANCOLOUR.pdf

http://marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-HEALTH-CHL-BLKSEA-OCEANCOLOUR.pdf

http://marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-HEALTH-CHL-BALTIC-OCEANCOLOUR.pdf

https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OC-QUID-009-141to144-151to154.pdf

https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OC-SQO-009-133to134.pdf

https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-OC-PUM.pdf